

# Zhihao Wang

Postdoctoral Researcher, University of Maryland

World Models | Scientific Machine Learning | Geospatial Computer Vision

zhwang0.github.io | Google Scholar | GitHub | LinkedIn

## Education

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|--------------------------------------|------------------------------------------------------|-----------|
| University of Maryland, College Park | Ph.D., Geographical Sciences                         | 2020–2026 |
| The Ohio State University            | M.A., Geography                                      | 2018–2020 |
| University of Waterloo               | B.E.S., Honours Geomatics; Minor in Computer Science | 2016–2018 |
| Wuhan University                     | B.E., Remote Sensing Science and Technology          | 2014–2018 |

## Research Experience

**First-author publications:** NeurIPS (2, Datasets and Benchmarks), AAAI (1), and ACM SIGSPATIAL (2).

**University of Maryland, College Park** 2020–Present

*Postdoctoral Researcher (Jun 2026–Present); Doctoral Researcher (2020–2026)*

**World Models & Scientific Machine Learning** NeurIPS’25; SIGSPATIAL’23, ’26

- Developed an Earth-system world model for **interactive what-if simulations**, enabling users to edit ecosystem states and explore coupled responses; learned from simulator trajectories without manually annotated interventions and evaluated across six global regions.
- Built **Deep-ED** to accelerate the EDv3.0 process-based model, combining multi-scale temporal modeling and knowledge-guided partitioning to **reduce long-horizon error accumulation by ~62%**.
- Created **CarbonGlobe**, the first global ML-ready dataset for forest carbon dynamics, to enable standardized benchmarking of **land-system emulation**; released 40 years of data at 0.5° resolution and evaluation scenarios for long-horizon generalization under climate shift. *Code* | *Data*

**Foundation Model Adaptation & Cost-Aware Evaluation** 2024–2026

- Developed a **cost-aware framework** across **13 models and 8 datasets**, translating prediction errors into estimated reference-data costs at fixed statistical precision; revealed why F1 alone misses cost differences driven by precision–recall trade-offs and class prevalence, informing model selection for real-world applications.
- Showed how **early stopping and checkpoint selection** can mask geospatial foundation models’ capabilities and change model rankings; compared three fine-tuning settings, with extended training and final-checkpoint evaluation improving mean macro-F1 by up to **2.3 percentage points** over early stopping.

**Computer Vision & Multimodal Model Routing** NeurIPS’25; ICDM Workshops’25

- Developed **TreeFinder** to bridge the gap between coarse satellite monitoring and local drone surveys: an open segmentation benchmark covering **48 U.S. states and 23,000+ hectares** at 0.6 m for individual-tree mortality detection and geographic generalization. *Code* | *Data*
- Mentored an intern** to develop **MARS**, using multimodal foundation models to automatically select suitable segmentation experts and address uneven performance across geographic scenes.
- Produced 10-year deforestation maps from approximately **15 TB** of 4.7 m satellite imagery to assess fine-scale forest loss beyond the detail of conventional monitoring products (e.g., 30 m or 10 m).

**Physics-Guided Fairness Learning** AAAI’24

- Developed **SimFair** to address uneven geographic performance under distribution shift without target-region training labels; used physics-guided inverse modeling to reduce **error disparity by 73%** on average across nine spatial splits versus a feedforward baseline.

**National Oceanic and Atmospheric Administration (NOAA)** May 2021–Aug 2022

*Graduate Research Assistant*

- Built climatology for cloud-gap filling in the **operational** NOAA-20 VIIRS Global Albedo product; processed **3.5+ TB** of data with Python multiprocessing on a Linux cluster.

## Selected Publications

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**First-author papers** | Full publication list: Google Scholar

- **Wang, Z.**, et al. Earth System World Model for What-If Simulations: A Case Study for Terrestrial Ecosystems. *ACM SIGSPATIAL 2026*.
- **Wang, Z.**, et al. CarbonGlobe: A Global-Scale, Multi-Decade Dataset and Benchmark for Carbon Forecasting in Forest Ecosystems. *NeurIPS 2025, Datasets and Benchmarks Track*.
- **Wang, Z.**, et al. TreeFinder: A US-Scale Benchmark Dataset for Individual Tree Mortality Monitoring Using High-Resolution Aerial Imagery. *NeurIPS 2025, Datasets and Benchmarks Track*.
- **Wang, Z.**, et al. SimFair: Physics-Guided Fairness-Aware Learning with Simulation Models. *AAAI 2024*.
- **Wang, Z.**, et al. High-Fidelity Deep Approximation of Ecosystem Simulation over Long-Term at Large Scale. *ACM SIGSPATIAL 2023; Oral Presentation*.

**Selected collaborative papers**

- Li, R., **Wang, Z.**, et al. EcoDiffusion: Uncertainty-Aware Emulation of Ecosystem Processes with Conditional Diffusion for Long Sequences with Single-Step Initialization. *AAAI 2026*.
- Li, C., **Wang, Z.**, & Xie, Y. Multimodal Foundation Models as Router Models for High-Resolution Aerial Image Segmentation. *IEEE ICDM Workshops 2025*.

**Preprints & Manuscripts**

- **Wang, Z.**, et al. Re-Imagining Foundation Models for Earth Monitoring: From Larger Models and Benchmark Gains to Operational Value. *Manuscript in preparation, 2026*.
- Mai, G., Rao, J., Cao, Q., Liu, Z., **Wang, Z.**, et al. Towards Generative Interactive Geospatial World Models for Proactive and Scenario-Driven Geospatial Decision-Making. *Manuscript under review, 2026*.
- Xie, Y., **Wang, Z.**, et al. When are Foundation Models Effective? Understanding the Suitability for Pixel-Level Classification Using Multispectral Imagery. *arXiv:2404.11797, 2024*.

## Technical Skills

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**Programming:** Python, C/C++, SQL, MATLAB, R, JavaScript.

**ML frameworks & methods:** PyTorch, TensorFlow; world models, physics-guided learning, time-series forecasting, semantic segmentation, foundation-model evaluation, geographic generalization.

**Data & research tools:** Linux, parallel data processing, Git, Google Earth Engine, Google Cloud, ArcGIS.

## Selected Honors

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|-----------------------------------------------------------------------------------------------------------------------------|------|
| <b>First Place (tie)</b> , UCGIS Student Lightning Talk Award                                                               | 2026 |
| <b>Outstanding Graduate Assistant Award</b> , University of Maryland                                                        | 2026 |
| <b>Scholar Award (travel)</b> , NeurIPS                                                                                     | 2025 |
| <b>Student Scholarship (travel)</b> , AAAI                                                                                  | 2024 |
| <b>Excellence in Graduate Research Award (Second Place)</b> , University of Maryland<br>Department of Geographical Sciences | 2024 |
| <b>Travel Award</b> , ACM SIGSPATIAL GIS Cup                                                                                | 2023 |
| <b>Third Place</b> , ACM SIGSPATIAL GIS Cup                                                                                 | 2022 |

## Service & Research Leadership

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- Organizing committee: IEEE BigData Cup 2026: AI Emulation Challenge; ACM SIGSPATIAL GeoIndustry workshop (2025, 2026).
- Program committee/reviewer: NeurIPS 2026; KDD (2025, 2026); AAAI 2026; ACM SIGSPATIAL (2025, 2026).
- Mentored three student researchers (2024, 2025). Instructor for Python Programming and Geocomputing, University of Maryland (2025).